

LMAP — Liquid Media Access Protocol

A Peer-to-Peer Protocol for Media Distribution and Ownership

Harrison Kavanaugh · contact@liquidmediafoundation.org

Version 3.0 — June 2026

Abstract

A purely peer-to-peer protocol for film distribution would allow digital media ownership to exist independently of any single platform, studio, or streaming service. The current model of centralized streaming creates fundamental misalignments: platforms bear infrastructure costs proportional to viewership, creators receive opaque compensation, and consumers own nothing permanent. We propose a layered protocol in which each film exists as fungible tokens within a unified registry contract, with separate provision for copyright ownership. Token quantity determines usage rights — from personal viewing to theatrical exhibition — at thresholds configured per title. Encrypted content is stored on a content-addressed network and incentivized via an economic substrate operated on a dedicated Layer-2 chain. Access to decryption material is gated by current on-chain ownership at decryption time, evaluated by a threshold cryptographic network with stake-based sybil resistance rather than by any single party. We claim, and intend to demonstrate, that this model provides security equivalent to or stronger than legacy hardware-attested digital-rights management, on an open-protocol substrate that the latter structurally cannot match. Forward-compatible support for hardware-attested key wrapping exists in the protocol's design for licensing relationships predicated on legacy industry compliance frameworks; we expect those frameworks to evolve to recognize threshold cryptography as equivalent or stronger security over time, and the protocol's center of gravity is positioned accordingly. A network of user-operated nodes — Seeds — distributes content, inverting the conventional cost model: the network strengthens as viewership grows. Smart contracts automate royalty distribution on every transaction, including secondary sales where collectors can sell exhibition rights directly. A single protocol token — LMA — coordinates storage incentives, threshold-network participation, and protocol value capture via emission to network operators and burn from protocol fees. The protocol is governed by an independent foundation, structurally distinct from any commercial operator. We argue that liquid media — physical-grade ownership in liquid digital form — is achievable without compromising the sovereignty that makes the design worthwhile.

1. Introduction

The transition to streaming was marketed as disruption, but the underlying economics reveal a fundamentally broken model. When a film succeeds on a centralized platform, infrastructure costs increase proportionally — the platform is economically punished for having popular content. This creates incentives misaligned with the goal of connecting audiences with compelling stories.

The consumer experience has equally degraded. Digital purchases are platform-locked, creating fragmented collections scattered across services. When platforms sunset or lose licensing rights, purchased content disappears. Unlike physical media, there is no true ownership — only temporary access privileges revocable at the platform's discretion.

Creators face similar opacity. Residual payments flow through accounting systems designed to obscure rather than illuminate. The shift from theatrical and physical media sales to streaming has compressed compensation without providing comparable transparency or participation in secondary markets.

What is needed is a protocol-level solution addressing all constituencies simultaneously — analogous to how DVD or USB function: not as a platform a consumer signs up for, but as a *standard* that many implementations interoperate through. We propose LMAP (the Liquid Media Access Protocol): a protocol where digital media is tokenized into standardized units granting verifiable ownership and access rights; content is distributed through a peer-to-peer network that scales positively with demand; smart contracts ensure automatic royalty distribution on every transaction; and ownership is structurally separable from any particular platform that runs on the protocol.

What consumers experience is *liquid media* — a generic public term for the category LMAP enables. Liquid media is to streaming services what physical media was to broadcast television: a different form of ownership, with different rights, that buyers can hold, transfer, and pass on.

2. Token Architecture

Each film exists as a unique token ID within a master ERC-1155¹ registry contract. Unlike ERC-721² non-fungible tokens, where each token is distinct, ERC-1155 allows millions of identical tokens per film, enabling both mass distribution and liquidity. A single contract scales to thousands of films, each identified by a unique token ID, with configurable quantities per title.

The token's URI points to a content identifier on a distributed storage network containing immutable metadata: title, synopsis, credits, and references to media assets. This architecture ensures that even if any particular implementation of the protocol ceases operations, token holders retain permanent access through the decentralized network and the open protocol specification.

Implementation status: shipped on Polygon mainnet as the `WyllohRegistryProtocolV4_1` contract at `0x8e834c6031A2d59e1cADd24Ab24F668301ed7c1D`. Verified source available via public block explorers. The V5 reference contract scaffold is under active development; deployment under foundation governance on the forthcoming Liquid Media Chain (§6) is a 2026 milestone.

3. Separation of Distribution and Copyright

A critical distinction exists between distribution rights and underlying copyright. The ERC-1155 registry manages distribution tokens — fungible units representing viewing and exhibition rights. A separate ERC-721 contract is specified to establish copyright ownership: a single non-fungible token per film representing the intellectual property itself.

The copyright token grants control over derivative works, merchandising, and the ability to propose governance actions such as token splits. Distribution token holders own rights to view, exhibit, and trade — but not to create sequels or merchandise. Crucially, the copyright holder cannot unilaterally mint additional distribution tokens; supply modifications require token holder approval, protecting holders from dilution. This mirrors traditional film finance, where investors hold revenue participation without controlling the underlying intellectual property, while adding protections absent in conventional structures.

This separation enables powerful scenarios. A studio could acquire the copyright token to produce a sequel while existing distribution tokens continue trading. The original filmmaker benefits from the sale; distribution token holders retain their viewing rights and market participation. Rights flow to natural owners without forcing all-or-nothing transactions.

Implementation status: design intent. The current V4.1 registry is the ERC-1155 distribution layer; the ERC-721 copyright registry is specified for a future protocol version (V5+) and has not yet been deployed.

4. Modular Rights Through Stacking

Token quantity determines usage rights through a threshold-based licensing system. A single token grants personal viewing rights — analogous to owning physical media. As tokens accumulate in a wallet, additional rights unlock at thresholds the filmmaker configures at tokenization time.

Thresholds are intentionally not fixed by the protocol; each filmmaker sets thresholds appropriate to their content's economics. A documentary might set low thresholds to encourage screenings; a genre release might set high thresholds reflecting commercial value. The protocol does not impose a universal price for theatrical rights because no such universal price exists in reality. The market discovers appropriate pricing through trading at the thresholds the filmmaker has chosen.

For illustrative purposes only — not as protocol-level constants — typical configurations might include:

- *1 token* — personal viewing rights
- *100 tokens* — small-venue exhibition
- *60,000 tokens* — theatrical exhibition rights with access to professional distribution formats

But these are choices, not specifications. Thresholds for any given film are encoded in that film's metadata at minting and are immutable thereafter for the life of that token's contract.

This mechanism creates natural price discovery for commercial rights. Rather than negotiating complex licensing agreements, distributors acquire sufficient tokens on the open market. Token supply is fixed at initial minting; if market conditions warrant increased liquidity, token holders may vote to approve a split, multiplying token quantities while preserving proportional ownership. The copyright holder may propose splits but cannot execute them unilaterally. New creative works — sequels, adaptations, alternative cuts — are tokenized as separate token IDs with independent supplies, leaving original token markets undisturbed.

Implementation status: shipped. Stacking thresholds and on-chain enforcement are operational in V4.1.

5. Layered Architecture

The protocol decomposes into eight layers. Each layer has a defined responsibility, a published interface to the layers above and below it, and an implementation status. Different parties can build different layers as long as the interfaces are honored. Conflating layers — particularly the entitlement layer with the cryptographic layer — is how comparable projects have failed.

#	Layer	Responsibility
0	Trust Anchors	Cryptographic primitives; hardware roots; wallet identity; contract immutability
1	Storage	Content distribution substrate (IPFS, manifests, incentivized pinning)
2	Cryptography	Encryption, key hierarchy, threshold release, watermarking
3	Entitlement	Smart contracts: distribution + copyright registries, royalties, rights tiers, payment splitters
4	Network	Peer discovery, shard request/response, LAN streaming
5	Forward Compatibility	Optional hardware-attested key wrapping for legacy industry licensing frameworks
6	Application	Storefronts, library UI, playback clients, integrator tooling
7	Governance	Foundation structure; federated certification; protocol governance

A storefront (Layer 6) only needs to read entitlement (Layer 3); a Seed manufacturer needs to implement Layers 1, 2, 4, and optionally 5; an academic auditor of the hardware-attestation framework only needs to engage with Layer 5 and the cryptographic claims of Layer 2. Each layer should be specifiable, testable, and replaceable independently of the others. The protocol does not mandate that any single party implement all eight layers.

The remaining sections of this paper traverse these layers from the chain upward, beginning with the chain that serves as the operational substrate.

6. The Liquid Media Chain

LMA operates on a dedicated Layer-2 chain — the *Liquid Media Chain* — stewarded by the Liquid Media Foundation (§16). The chain provides the execution substrate for LMA smart contracts (Layer 3), the economic substrate for the LMA token (§16), and the operational substrate for sequencer infrastructure that the protocol’s contracts settle through.

The chain is constructed as a zero-knowledge rollup³ built on the Polygon CDK framework, settling to Polygon mainnet as its Layer-1 security anchor. Polygon CDK was selected over alternative rollup frameworks for three reasons: it is production-grade with established managed-service operators (Conduit, Caldera, Polygon Labs itself), reducing foundation operational burden during bootstrap; its zero-knowledge security model provides fast finality without fraud-proof challenge windows, improving user experience for purchase and transfer flows; and settlement on Polygon preserves USDC liquidity and the existing V4.1 contract investment, enabling holders of legacy tokens to bridge into the chain without disruption.

The chain’s block economics differ from those of general-purpose Layer-2 chains in three ways. First, the chain uses LMA as its native gas asset, giving the LMA token operational utility tied directly to protocol usage. Account abstraction⁴ paymaster contracts allow users to pay gas in USDC where preferred, insulating consumers from gas-asset volatility; LMA’s role as native gas is structural to the protocol’s incentive design rather than a user-facing surface. Second, a configurable percentage of every transaction’s gas fee is burned (initial parameter: 50%),

contracting LMA supply in proportion to chain usage. This follows the pattern established by Ethereum’s EIP-1559⁵ base-fee burn, applied to a media-specific chain. Third, a fraction of emission per block flows to storage providers that successfully prove retrievability of registered film CIDs in that epoch (§7), coupling the storage incentive directly to chain operations rather than running as a separate off-chain mechanism.

Bridges connect the Liquid Media Chain to Polygon mainnet for USDC, POL where required, and any V4.1 token holdings. Bridge contracts are audited and operated by the Liquid Media Foundation during bootstrap, with the option to migrate to a decentralized bridge protocol at network maturity. USDC bridged from Polygon retains its peg and serves as the canonical commerce currency on the Liquid Media Chain: filmmakers receive USDC; storefront markup flows in USDC; the protocol fee accumulates in USDC at the foundation treasury for the periodic burn auctions specified in §11.

Implementation status: specification. The chain is in design; its launch under foundation governance is a Year-1 milestone of the protocol’s bootstrap roadmap (§19). Until launch, LMAP operates on Polygon mainnet as the bootstrap substrate. The V4.1 contracts continue to serve existing tokens through and beyond the chain’s deployment.

7. Content Storage and Incentivized Pinning

Media files are encrypted using AES-256-GCM before distribution on a content-addressed storage network⁶. The metadata structure nests content identifiers: the token URI points to a metadata file containing references to both unencrypted promotional assets (poster, trailer) and encrypted media files. Each encrypted asset specifies a threshold — the token quantity required for access — corresponding to the rights tiers established at minting.

Encrypted content is stored in a chunked format (`[Length(4 bytes)][IV(12 bytes)][Ciphertext+Tag(16 bytes)]` per 4-MiB chunk) which is self-authenticating per chunk and supports streaming decryption with constant memory. The format is documented in the protocol specification; any compliant client can implement encryption and decryption against it.

7.1 Storage Providers and Proof of Retrievability

Any participant operating an IPFS daemon plus the LMAP storage-provider module can register as a storage provider. Registration requires staking LMA at a storage-registry contract on the Liquid Media Chain. Wylloh’s Seed One and Origin Seed hardware (§10) are storage-provider-capable by default; software-only nodes operating on commodity Linux servers or virtual machines may also participate on identical protocol terms.

Storage providers earn LMA per epoch based on proof of retrievability: the protocol periodically challenges providers with random byte-range requests; providers must respond within a configured deadline with the requested bytes plus a cryptographic proof binding the response to the original CID. Failure to respond, or response with incorrect bytes, triggers slashing of staked LMA proportional to the severity. Reward weighting incorporates latency to first byte, geographic distribution bonuses (encouraging spread across regions with low density), and sustained-pinning consistency. The design borrows from Filecoin⁷ proof-of-replication and proof-of-spacetime mechanisms, adapted for streaming-media chunks rather than arbitrary archival data: more frequent small challenges rather than infrequent large proofs, and incorporation of latency and geographic factors that matter for media delivery.

7.2 Threshold Dispersal

A future enhancement at this layer is *threshold dispersal of ciphertext shards* — distributing encrypted content shards across the Seed network using Reed-Solomon erasure coding, with the option of secret-share-based reconstruction⁸, such that reconstruction requires a quorum of nodes. This is a resilience property at the storage layer, not the load-bearing security mechanism (see §8.4 for the precise scope of this claim).

Implementation status: chunked AES-256-GCM encryption, IPFS storage, and a centralized pinning fallback are shipped. Stake-and-slash incentivized pinning, proof-of-retrievability, and the storage-registry contract are specified for deployment on the Liquid Media Chain in Year 1. Threshold ciphertext dispersal is specified for a subsequent protocol version.

8. Cryptography and Threshold-Based Access Control

LMA's security model is threshold-mediated key release. Access to a film's decryption material is gated by *current on-chain ownership of the corresponding token*, evaluated at decryption time by a threshold cryptographic network. No party — not the foundation, not a storefront, not a hardware vendor, not the protocol's own contracts — can release decryption material to a wallet that does not currently satisfy the access condition.

We claim, and intend to demonstrate, that this model provides security equivalent to or stronger than legacy hardware-attested DRM systems, while preserving open-protocol participation in ways those systems structurally cannot. Threshold cryptography with stake-based sybil resistance is mathematically sound and has not been broken in production deployments; Widevine L1, by contrast, has been broken multiple times. The protocol does not require hardware attestation to be secure; it requires honest cryptography and aligned economic incentives, both of which the threshold model provides.

8.1 A Note on the Legacy Open-Tier Approach

Earlier protocol versions described an *open tier* in which the content-decryption wrapping key was deterministically derivable from public on-chain data using a documented formula. Decryption then proceeded entirely with public inputs: a wallet that had once received the encrypted master key could decrypt forever, regardless of subsequent token transfers.

This approach was articulated as a deliberate design choice in service of platform independence. Reviewed against the requirements of a production protocol for film distribution — independent or otherwise — it has been retired. The derivable-key construction does not gate decryption by current ownership: any party with the public chain data and the published metadata can derive the wrapping key, retrieve the encrypted master key from public storage, and decrypt the film without ever holding the token. The construction is therefore appropriate for *demonstrating the protocol's other mechanics on public-domain content* — the role it played in the V4.1 deployment with the 1929 public-domain film *The Cocoanuts* — and is not appropriate for any content of commercial value.

V3 of the protocol replaces the open-tier construction with threshold-mediated key release as the standard production mechanism. Legacy V4.1 tokens continue to operate under the original construction for historical continuity; no new commercial content is tokenized against the legacy construction.

8.2 Threshold-Mediated Key Release

Threshold cryptography enables a network of N nodes to collectively release a master key only when a configurable threshold T of them independently verifies an access condition. No single node — and no group fewer than T — can release the key. The mathematical security depends on honest-majority assumptions; the operational security depends on stake-based participation, in which threshold-network operators stake LMA and have stake slashed for misbehavior.

At content preparation, a publisher generates a random master key and encrypts the film with it using chunked AES-256-GCM. The publisher then submits the master key to the threshold network for encryption to a Distributed Key Generation public key. The network returns a threshold-encrypted master key, which is stored alongside the metadata. The film's metadata declares an Access Control Condition — typically `balanceOf(:userAddress, :tokenId) > 0` against the LMA registry contract.

At decryption, a client signs a SIWE-style authentication message⁷ proving control of the requesting wallet. The client submits the authentication, the threshold-encrypted master key, and the Access Control Condition to the threshold network. Threshold-network nodes independently verify the wallet signature, query the chain for current Access Control Condition state, and release decryption shares only if the condition evaluates true. The client collects T shares and reassembles the master key, then decrypts the film using the standard chunked AES-GCM mechanism.

Transfer behavior emerges from the architecture. When a token is transferred from one wallet to another, the Access Control Condition is evaluated against current chain state at the next decryption attempt. The seller's wallet no longer satisfies `balanceOf > 0`; the buyer's now does. The threshold network refuses to release shares to the seller and releases them to the buyer. No re-wrapping, no foundation involvement, no seller cooperation required.

8.3 Year-0 Bootstrap on an Established Threshold Network

During bootstrap, LMAP integrates with Lit Protocol's Naga mainnet¹⁰ as the threshold-network substrate. Lit is a production-grade threshold-decryption network with established support for chain-based Access Control Conditions, including Polygon, and several years of operational maturity. Year-0 commercial content is tokenized against Lit using the threshold-mediated mechanism.

Beginning in Year 1, the Liquid Media Foundation deploys an LMAP-native threshold network operated by registered threshold-network nodes — initially foundation-operated bootstrap nodes, subsequently Wylloh Seeds and any third-party operator that stakes the required LMA minimum. New content increasingly defaults to the LMAP-native network through Year 1. By the end of Year 2, all production content has migrated to the LMAP-native threshold network; the Lit Protocol dependency is retired.

The bootstrap dependency on Lit is transient and intentional, not permanent reliance. Replacing Lit with the LMAP-native threshold network is a public protocol commitment with a defined milestone, not a discretionary path.

8.4 Forward Compatibility for Legacy Industry Frameworks

Some industry licensing frameworks — primarily those originating with major studios — reference specific hardware-DRM technologies (Widevine L1, PlayReady, FairPlay) as contractual conditions of licensing. These frameworks predate threshold cryptography's deployment maturity and were written when hardware-attested per-device key custody was the only practical mechanism for cryptographic enforcement at scale. We anticipate that these frameworks will evolve to recognize threshold cryptography as equivalent or stronger security over time; this is a reasonable expectation given threshold cryptography's production track record and the existing frameworks' demonstrated breakability.

In the interim, LMAP's design includes forward-compatible support for hardware-attested key wrapping. This support is a capability of the protocol, not a positioning choice or a marketing feature. The protocol's primary security guarantee runs through threshold release; the hardware-attested capability exists so that licensing relationships predicated on the legacy frameworks can be honored without forking the protocol or compromising its open-protocol participation properties.

When invoked for a specific content class whose licensing contract requires it, the hardware-attested capability operates as follows:

- Each participating Seed holds a per-device keypair generated inside its hardware secure element. The private key is non-extractable.
- For attestation-required content, the threshold network's released master key is additionally wrapped to the Seed's per-device public key. Unwrapping occurs only inside the secure element; the unwrapped key never appears in main memory or persistent storage outside the secure perimeter.
- Wrapped-key issuance requires the Seed to present a fresh *attestation report* signed by its secure element, containing current firmware measurements. The report is verified against the Seed's attestation credential before

any wrapped key is issued.

- Per-title content keys are derived from a master with HKDF-style domain-separated key derivation¹¹ to prevent key reuse across titles.

Reference hardware for attestation-required operation includes ARM TrustZone¹²-capable SoCs with discrete secure elements (e.g., NXP SE050 or equivalent), and platforms with TPM 2.0¹³ support. Remote attestation¹⁴ follows established patterns from contemporary trusted-computing literature.

Hardware binding is a normative requirement when the hardware-attested capability is invoked. A compliant implementation of attestation-required content delivery MUST use a hardware secure element to (a) generate the per-device keypair, (b) hold the private key in non-extractable form, (c) perform key unwrapping, and (d) sign attestation reports. An implementation that performs any of these operations outside a hardware secure element is not conformant.

A clarifying distinction: the hardware-attested capability gates *decryption*, not *carriage*. Encrypted content under this capability is freely distributable through the same content-addressed substrate as all other LMAP content; any device can hold, pin, or serve the encrypted bytes. Tokens remain freely transferable across all wallets. Hardware attestation, when invoked, gates playback only. This preserves user sovereignty (a buyer's ownership is unaffected by hardware availability; they can transfer their token at any time) while honoring the legacy contractual requirement (decryption is impossible without an attested secure element).

The protocol's posture is explicit: this capability is forward-compatible for industry frameworks that have not yet evolved. As licensing counterparties update their compliance frameworks to recognize threshold cryptography directly, the relevance of the hardware-attested capability decreases. The protocol's intellectual center of gravity is threshold-mediated release; the hardware-attested capability exists at the periphery, as a courtesy to a legacy that is itself in the process of becoming obsolete.

8.5 Federated Issuance for the Hardware-Attested Capability

When the hardware-attested capability is invoked, attestation credentials are signed by independent *attestation issuers*. An issuer verifies a Seed's hardware integrity and signs a credential binding the device's public key to a manifest of approved firmware measurements. Issuers operate under a federated framework governed by the Liquid Media Foundation (§16).

No single issuer can unilaterally approve or deny attestation; the framework's governance can revoke any issuer whose practices fail audit. Counterparties negotiate with the framework, not with a single platform operator. The model is structurally analogous to the federation of certificate authorities that underpins the public web's TLS infrastructure: independent issuers, audited practices, root trust managed by a body distinct from any commercial participant.

The framework launches with a single issuer (the Liquid Media Foundation itself) and a public commitment to federate at a defined milestone of network maturity. Federation is a visible commitment, not an indefinite promise. We note that the framework's relevance itself depends on the persistence of the legacy industry requirements that necessitated it; over time, both the issuer federation and the hardware-attested capability are expected to decrease in significance relative to threshold-mediated release.

8.6 Security That Scales with Network Size

Two distinct claims about how protocol security relates to network size deserve separate treatment, because they have different defensibility.

Strong claim — threshold release with stake-based sybil resistance. Compromising the threshold network requires either compromising more than the slashing-tolerance threshold of independently-staked nodes (economically irrational at scale) or extracting cryptographic material from those nodes (each holds only a share).

The network’s aggregate exposure does not grow with the network’s size; if anything, it shrinks, because additional staking pools dilute the share that any attacker would need to control. This is the load-bearing cryptographic claim.

Strong claim — per-device key wrapping (hardware-attested capability, when invoked). When the hardware-attested capability is invoked, compromising one attested Seed yields that Seed’s local content only. The network’s aggregate exposure does not grow with the network’s size, because each attested device is its own isolated vault. This is a property of the capability when invoked, not a property the protocol depends on for its primary security guarantee.

Modest claim — threshold dispersal of ciphertext (storage layer). Sharding ciphertext across Seeds raises the per-shard extraction cost. An attacker compromising one Seed obtains a fraction of the ciphertext for any given film; reconstruction requires reaching the quorum threshold. This is real but bounded: a sufficiently motivated attacker can target enough Seeds to reach quorum. The combinatorial cost is meaningful but does not produce asymptotic security.

This precision matters for technical readers because the conventional DRM model degrades as user base grows (more devices, more keys, more attack surface). Threshold release and per-device key wrapping invert that property; threshold dispersal at the storage layer supplements it.

8.7 Watermarking

Forensic watermarking binds a leak to a specific wallet, raising the cost of unauthorized distribution beyond the bare cryptographic extraction cost. Watermarks may be inserted server-side at content-preparation time (per-purchase variants delivered to specific wallets) or, for attested-mode content, Seed-side at decryption time, inside the secure element, before plaintext leaves the secure perimeter.

Watermark robustness against transcoding and adversarial filtering is an ongoing research area; commercial watermarking implementations exist with established studio relationships. The protocol specifies watermark insertion points but does not mandate a particular scheme.

Implementation status: threshold-mediated key release against Lit Protocol’s Naga mainnet is in active implementation; migration of the Wylloh reference client and Seed daemon from the legacy derivable-key construction to threshold-mediated release is a near-term engineering milestone, completed before any commercial title is tokenized. The LMAP-native threshold network is specified for Year-1 deployment on the Liquid Media Chain. Hardware-attested mode is specified at the protocol level with hardware-binding declared normative (§8.4); the wire-level attestation specification, the framework’s first issuer beyond the foundation, governance process, and operational maturity are forthcoming. Server-side watermarking is in initial implementation with a commercial watermarking provider relationship to be established before commercial-content launch.

9. Permanence Guarantees

The protocol commits to a permanence guarantee: a buyer who has legitimately acquired a film at any point retains the ability to play that film indefinitely, regardless of subsequent network availability, attestation revocation, or platform fate.

The guarantee is engineered through three mechanisms operating in combination. First, after a client has successfully decrypted a film through the threshold network at acquisition, the master key is cached locally — in the browser’s persistent storage, the Seed daemon’s encrypted local storage, or any compliant client implementation’s local persistence. Subsequent playback of the same film does not contact the threshold network; the client decrypts directly from local state. Second, holders may export their master keys as user-encrypted backups — wrapped with a passphrase the holder chooses, written to any storage the holder controls. A backup file plus the downloaded encrypted bytes plus the user’s passphrase enables playback in perpetuity, even if all foundation and threshold-

network infrastructure ceases to exist. Third, the ownership record on the chain remains the durable source of truth; a holder who has lost both local cache and backup can re-acquire access through any compliant implementation of the protocol that still operates against the chain, because the wallet's holdings are recoverable from public data.

In attested mode, the permanence guarantee is preserved by a parallel mechanism. Attestation gates the *issuance* of wrapped keys, not the *playback* of content already keyed. A Seed whose attestation credential is later revoked retains the wrapped keys it previously received and continues to play the corresponding films. Revocation prevents the Seed from acquiring further content through attested mode; it does not retroactively disable the Seed's existing library. This is the architectural answer to the rugpull pattern that has characterized streaming services and digital media stores.

The hardware itself remains a single point of failure: a Seed whose secure element fails physically loses access to its locally-wrapped keys. The protocol mitigates this by treating the on-chain ownership record as the durable source of truth. A user whose Seed has died can re-acquire wrapped keys for their owned films onto a new attested Seed, because the wallet's holdings are recoverable from the blockchain. **The blockchain remembers; the hardware is replaceable.** This is a stronger guarantee than physical media, which has no equivalent recovery path when a disc fails.

The canonical claim, stated precisely: *downloaded films play forever offline; re-download and transfer require an access network — the Lit Protocol Naga mainnet during Year 0; the LMAP-native threshold network from Year 1 onward — to be live; no single-company point of failure.*

Implementation status: local master-key caching, the self-wrapped backup export, and the recovery path through on-chain ownership are in active implementation. The Year-0 threshold-network dependency on Lit Protocol is acknowledged honestly in user-facing communications; the Year-2 sunset of that dependency in favor of the LMAP-native network is a public commitment.

10. The Seed Network

Content distribution operates through *Seeds* — user-operated nodes that combine encrypted-content storage, peer-to-peer participation, a local LAN-streaming service for playback applications, and (when staked) threshold-network operation and storage-incentive participation.

A Seed is a headless device. It does not connect directly to a television; instead, it serves LMAP-registered content to client applications running on platforms users already own — Roku, Apple TV, iOS, the web. The user's existing remote, the user's existing television, the user's existing input habits all remain intact. The Seed becomes infrastructure: quiet, reliable, operated as a piece of furniture in the home network rather than a screen in the living room.

Each Seed stores content its operator owns, plus optionally contributes overflow capacity to the network. Popular films accumulate across more Seeds as more users own tokens; niche content maintains availability proportional to its audience. No central coordination determines what Seeds store; ownership patterns drive distribution.

This architecture inverts conventional streaming economics. Centralized platforms bear infrastructure costs proportional to viewership: success is expensive. In the Seed network, increased viewership means more Seeds joining, distributing load across more nodes. Per-stream costs approach zero as the network grows. The system becomes more efficient at scale rather than more costly. Seeds that stake the LMA minimum (§16) participate in the storage-incentive network (§7) and the threshold network (§8); they earn LMA emission for proof-of-retrievability work and threshold-share-release work. Network participation is therefore both a beneficial side effect of a device that operators would buy regardless (the local film library is the primary purchase rationale) and an ongoing source of compensation for operators who choose to stake.

10.1 Reference Hardware

Two reference Seed devices are specified by the protocol's first commercial implementation (§17). The *Seed One* is a compact, fanless, consumer-priced reference device specified to participate fully in the protocol on commodity hardware: a current-generation single-board computer, a hardware secure element, network storage, and a published reference firmware. The *Origin Seed* is a sculptural, hand-finished reference device specified at the high end of the product line: substantially greater storage capacity, audiophile-grade power supply, and a furniture-quality enclosure intended for the luxury custom-integration market.

Both reference devices participate identically at the protocol level. Both are storage-provider-capable, threshold-network-operator-capable, and hardware-attestation-capable. The distinction between them is form, capacity, channel, and price — not protocol participation.

The protocol does not require either reference device. Any compliant Seed — third-party hardware, software-only nodes operating on commodity Linux, virtual-machine instances — can participate on identical terms. The reference Seeds exist as canonical implementations the protocol can point to; they are not the only implementations welcomed.

10.2 Operational Behavior

A compliant Seed daemon verifies wallet ownership directly against the on-chain registry, fetches encrypted content from IPFS (preferring a co-resident node where present, falling back to public gateways), decrypts streaming with constant memory, automatically pins source CIDs to make the Seed a provider for content it holds, and exposes a long-running HTTP service with mDNS auto-discovery and HTTP byte-range streaming for LAN client applications. Where the Seed has been configured for it, the daemon additionally participates in the threshold network as a stake-bonded node, releasing decryption shares for content the network requests in accordance with §8.

Implementation status: an open-source reference Seed daemon is operational, demonstrating wallet-verified content fetch, decryption, and LAN streaming on independent ARM64 hardware (Raspberry Pi 4) with byte-identical decrypted output across architectures and no dependency on any centralized service. The reference Seed daemon's migration from the legacy derivable-key construction to threshold-mediated release is a near-term milestone. Both reference Seeds — *Seed One* and *Origin Seed* — are in pre-production. The web client at wylloh.com serves as the first reference application.

11. Marketplace Mechanics

The protocol includes integrated marketplace functionality. Primary sales occur through minting at prices set by the title's author. Secondary sales enable peer-to-peer trading with automatic royalty distribution on each transaction. All marketplace operations execute against smart contracts on the Liquid Media Chain (§6); commerce settles in USDC with gas paid in LMA underneath an account-abstraction paymaster so consumers experience USDC-only checkout.

11.1 Three-Way Revenue Split

Every primary sale routed through the protocol distributes payment across three parties, with proportions set per title at mint time:

- **Protocol fee: 2.5%** (immutable, hardcoded at the contract level). Flows to the Liquid Media Foundation treasury in USDC. The foundation retains a small operating carve-out (initial parameter: 10% of accumulated fee revenue) for shared infrastructure: IPFS pinning redundancy, contract auditing, security work, open-source maintenance. The remainder flows into the periodic burn auctions described in §11.8.
- **Publisher fee: 0–25%** (set per title, immutable for that title). Flows to the publisher — the wallet that minted the title. Publishers are entities that bring works to market: established storefronts, indie collectives, or self-publishing authors. Their cut compensates curation, marketing, audience development, and brand value.

- **Author share: remainder.** Flows to the author — the credited creator(s). Distributed via an optional royalty shareholders configuration (§11.3) supporting up to 50 collaborators per title.

For a typical configuration — 10% publisher fee — the split is 2.5% / 10% / 87.5%. Authors receive dramatically better economics than traditional publishers, where author shares of 10–20% are common. The protocol’s 25% publisher cap is the only opinionated stance: *publishing on LMAP shall not become as extractive as the systems it replaces.*

11.2 Publisher and Author as Distinct Roles

The protocol records two distinct identities per title:

- **Publisher** is set automatically from the wallet that calls the mint function. Cryptographically authentic — cannot be impersonated. Establishes the brand identity associated with the title and receives the publisher fee on primary sales.
- **Author** is provided at mint and identifies the credited creator of the work. Receives the author share of every primary and secondary sale.

The two roles are often the same wallet (an author who self-publishes) but can differ (a publisher mints under their brand and routes the author share to a separate creator wallet). This mirrors the publishing-house / author distinction familiar from books, films, and music.

11.3 Royalty Shareholders

Authors can configure up to 50 royalty shareholders per title, each receiving a configurable percentage of the author’s share. This supports the realistic case where a tokenized work has multiple contributors deserving compensation: a film with director, lead actors, composer, screenwriter; a video game with developers, artists, and key team members; an album with songwriter, performers, and producer.

Shareholders are mutable by the author, allowing collaborators to be added or adjusted as the work evolves. Sum of shares can be $\leq 100\%$ of the author’s share; any remainder flows to the author wallet, preserving the author as the residual recipient. For projects with more than 50 beneficiaries, shareholders compose with external splitter contracts: one shareholder slot can route to a splitter address that distributes further.

Distribution happens automatically on every protocol-routed sale — both primary and secondary. No manual royalty disbursement, no escrow, no off-chain ledger reconciliation.

11.4 Royalty Enforcement Boundary

The protocol enforces royalties cryptographically for sales routed through its marketplace functions. External marketplaces honoring the ERC-2981 royalty standard will respect the configured royalty when displaying and processing sales. Direct wallet-to-wallet transfers — the gift, inheritance, and personal-give use cases — do not trigger royalties; this is a property of the underlying ERC-1155 standard and a deliberate preservation of token-as-property semantics.

Off-chain payment with on-chain gift is theoretically possible and unenforceable cryptographically, as with every royalty-bearing token system. The protocol’s mitigation is structural: the low 2.5% protocol fee minimizes the savings from circumvention; the integrated marketplace makes legitimate sales easier than coordination; on-chain transparency makes wash-trade patterns visible and reputationally costly.

11.5 Bulk Acquisition for Rights Stacking

The marketplace supports atomic batch purchases, enabling a single transaction to acquire tokens from many sellers simultaneously. This is essential for the rights-stacking use case (§12): a buyer assembling tokens to meet a theatrical-exhibition threshold may need to acquire thousands of tokens from hundreds of sellers in a single

coordinated purchase. The atomic semantics guarantee all-or-nothing: if any listing in the batch fails, the entire transaction reverts, preventing the partial-fill case where a buyer pays fees but does not reach their target threshold.

Platform UIs aggregate, sort, and present available listings; the protocol provides the batch-execution primitive.

11.6 The Protocol Fee Is Not a Platform Fee

The 2.5% protocol fee supports shared infrastructure all implementations depend on, identical across every LMAP-compatible marketplace. Third-party marketplaces may layer their own commercial markup on top, competing on user experience, curation, and discovery rather than by undercutting the protocol fee. This places third-party marketplaces and the reference web client at [wylloh.com](https://www.wylloh.com) in a peer relationship with one another, not a competitive one.

11.7 Settlement and Gas Abstraction

Transactions settle in stablecoin to insulate buyers from cryptocurrency volatility. On the Liquid Media Chain (\$6), settlement is in USDC; gas is paid in LMA underneath an ERC-4337¹⁵ paymaster contract such that users experience USDC-only checkout. The paymaster contract pays LMA gas on the user's behalf, recovering the cost via a small USDC surcharge bundled into the transaction. Paymasters are deployed and funded at the marketplace layer; each marketplace can choose whether and how to subsidize gas as part of its commercial offering.

The legacy V4.1 deployment on Polygon mainnet operates against USDC.e and requires users to hold POL for gas — friction the protocol acknowledges as a near-term limitation of the bootstrap substrate rather than an intended feature. Migration to the Liquid Media Chain removes both: USDC replaces USDC.e; LMA gas under the paymaster removes the user-facing POL requirement.

11.8 LMA Token Burn from Protocol Fees

The Liquid Media Foundation operates periodic Dutch auctions converting accumulated USDC protocol-fee revenue into LMA, which is then permanently burned. The auction cadence is configurable (initial parameter: weekly); the burn destination is the conventional zero-public-key burn address. Burn events emit on-chain logs; supply contraction is public and auditable.

This mechanism couples LMA supply contraction directly to protocol usage. Every sale on every LMAP-compatible marketplace contributes to the burn pool. As protocol usage grows, the burn rate grows; the LMA token captures value not through extraction by any intermediary, but through deflationary pressure proportional to actual network activity. The structure follows the precedent of Ethereum's EIP-1559¹⁶ base-fee burn, with the analogous result: token holders benefit from network growth proportionally, with no entity acting as an extractive operator.

Implementation status: primary sales with three-way distribution, secondary sales with royalty enforcement, royalty shareholder distribution, batch acquisition, and the protocol fee are specified in the V5 reference contract. The V5 contract is in active development; deployment under foundation governance on the Liquid Media Chain is a 2026 milestone. The legacy V4.1 registry remains in operation for existing tokens (legacy derivable-key construction, two-way author / protocol split) during the transition. The USDC-to-LMA burn auction is specified for the Liquid Media Chain deployment.

12. Commercial Rights Discovery

The stacking mechanism enables a novel form of commercial rights acquisition. Consider a film that achieves cult status: tokens are distributed across thousands of fans who purchased during initial release. A theater chain wishes to conduct a theatrical run, requiring tokens above the exhibition threshold the filmmaker configured for that title.

Rather than negotiating with the original filmmaker, the theater acquires tokens directly from fans through open market purchases. As buy pressure increases, token prices rise, rewarding early believers who funded the film's success. Royalties trigger on each transfer, flowing to the copyright holder regardless of who sells. The market determines that theatrical rights to this particular film, at this particular moment, command a specific price — no negotiation required.

Commercial exhibition requires tokens to be staked for the exhibition period. The professional distribution system verifies on-chain that staked tokens meet the exhibition threshold before enabling playback. This prevents the exploitation pattern of acquiring tokens, screening content, and immediately selling. Staking creates commitment aligned with legitimate commercial use.

Implementation status: design intent. Stacking thresholds are operational at the contract level today; the staking mechanism for commercial-exhibition windows is specified for a future protocol version.

13. Production Funding

The token architecture enables presale-based production financing. Rather than relying on studio executives or platform algorithms to greenlight projects, creators can presell tokens directly to audiences. This validates market demand before production begins while distributing ownership of the work directly to its audience.

Presale tokens grant the same ownership rights as post-release tokens but are purchased at a discount, rewarding early supporters. Smart contracts can structure milestone-based fund releases with multi-signature approval from stakeholder representatives. If production fails to meet milestones, remaining funds return to presale participants.

This model removes centralized gatekeepers from greenlighting decisions. A film need not fit a platform's content strategy. If sufficient audience interest exists to fund production, the project proceeds. The presale serves simultaneously as market research, financing mechanism, and community-building exercise.

Implementation status: design intent. The token architecture supports presales architecturally; the milestone-based escrow contracts and the corresponding presale-tokenization workflow are specified for a future protocol version.

14. Privacy

Contemporary streaming platforms extract extensive personal data to drive recommendation algorithms and advertising. The protocol operates on a different principle: *movies should not watch their viewers*. Analytics derive exclusively from public blockchain data — transaction patterns, token distribution, market activity — rather than viewing behavior or personal information.

The conventional streaming model links user identity to viewing history to platform to advertisers, forming a chain of surveillance maintained primarily for monetization through inferred consumer preferences. The protocol model maintains pseudonymous wallets executing public transactions on a transparent ledger, with no link to real-world identity, no telemetry from playback events, and no viewing patterns aggregated by any party. Creators receive aggregate insights — token distribution by region inferred from public data, secondary-market activity, holder concentration — without compromising individual privacy.

15. Security Philosophy

The protocol does not attempt perfect copy protection. History demonstrates that technological restrictions on digital media inevitably fail while degrading legitimate user experience — including the legacy hardware-DRM systems that this protocol claims to equal or exceed. Widevine L1 has been broken in production multiple times; PlayReady and FairPlay have had their own compromises. The premise that the user is an adversary, and that sufficient hardware can prevent extraction, has not held in practice.

Instead, the protocol aligns incentives such that ownership provides value beyond mere access — liquidity, permanence, community membership — making legitimate acquisition the rational choice. The cryptographic enforcement that *is* present (threshold-mediated key release, §8) gates access by current on-chain ownership, evaluated by a network of stake-bonded nodes. This provides honest cryptographic protection against the systematic extraction case without treating audiences as adversaries.

The analog hole — capture of content via camera or screen recording during legitimate playback — remains possible. Watermarking is the only mitigation, and watermark robustness against transcoding is an ongoing arms race. The protocol does not claim immunity from this class of attack; it claims that legitimate purchase confers economic benefits that bare ciphertext extraction does not, and that forensic watermarking enables attribution of leaks to specific purchasers.

When content is owned not by a distant corporation but by potentially many individual token holders, piracy dynamics shift. Unauthorized distribution harms not an abstract studio but a community of fellow enthusiasts. Token holders own copies of the work itself, aligning their interest with its integrity rather than its piracy. Social contracts, properly constructed, provide stronger protection than technological locks against the casual case; threshold-mediated key release provides cryptographic protection against the systematic extraction case. These two mechanisms together are the protocol’s security posture; they do not require — and do not depend on — hardware-attested per-device key custody. Where forward-compatible hardware-attested key wrapping is invoked for licensing relationships predicated on legacy industry frameworks (§8.4), it operates as an additional credential check, not as the load-bearing security layer.

Physical media operated on similar principles. Copy protection on optical discs was trivially circumvented, yet the market thrived for decades. Convenience, quality assurance, and social norms around ownership provided sufficient protection. The protocol adds financial incentives physical media lacked: ownership appreciates in genuinely successful work, and secondary markets reward early adoption.

A specific guidance for technical readers: when evaluating the protocol’s claims, separate the *cryptographic claim* (threshold-mediated release with stake-based sybil resistance provides security equivalent to or stronger than legacy hardware-attested DRM, and does not degrade with network scale) from the *operational claim* (legitimate purchase is more economically rational than extraction for participants in the network). Both claims hold, but they hold at different layers and under different threat models. The whitepaper is precise about this because the audiences this document serves — investors, developers, and licensing counterparties — read the same words and need to extract the same meaning.

16. Governance and the Liquid Media Foundation

The protocol is governed by an independent foundation, structurally distinct from any commercial operator. The Liquid Media Foundation holds the protocol’s intellectual property under a permissive open-source license, operates the Liquid Media Chain (§6) sequencer during bootstrap, manages the LMA token’s issuance and burn mechanics, stewards the forward-compatible attestation framework (§8.4) for licensing relationships predicated on legacy industry requirements, publishes the protocol specification, and operates as a protocol-neutral standards body.

Commercial entities that build on the protocol — storefronts, hardware manufacturers, marketplace operators — are participants in the LMAP ecosystem rather than its operators. This structural separation mirrors precedents in adjacent ecosystems: the Filecoin protocol governed by the Filecoin Foundation, structurally distinct from Protocol Labs; the Ethereum protocol governed independently from any single Ethereum company. The pattern preserves credible-neutrality: filmmakers, third-party storefronts, hardware manufacturers, and eventually studios can build on the protocol without fearing that the protocol’s stewards are also their commercial competitors.

16.1 Legal Structure

The Liquid Media Foundation is structured as a *Cayman Foundation Company*. This structure has established legal precedent for protocol stewardship under regulatory frameworks that recognize utility-token issuance — the Tezos Foundation, the Cardano Foundation, the Polkadot Web3 Foundation, the BNB Foundation, and others follow analogous structures. The Cayman Foundation Company offers three properties relevant to LMAP: cross-jurisdictional flexibility (the protocol’s participants are global), token-issuance precedent (the structure has been tested against securities-law analysis in adjacent contexts), and operational separation between the foundation’s protocol-stewardship role and any commercial participant in the ecosystem.

A pragmatic note about the present: the foundation’s legal structure is the working commitment for v3 of this whitepaper, made on informed-but-pre-counsel grounds. Formal incorporation and the securities-counsel review attendant to LMA token issuance are funded milestones of the protocol’s bootstrap roadmap (§19). If counsel review identifies a structurally superior alternative, v3.1 of this whitepaper will reflect the revised commitment; readers should treat §16.1 as the current working commitment, not as a ratified outcome of completed legal review.

16.2 What the Foundation Stewards

The foundation, at its mature governance, stewards:

- **Specification governance.** Protocol upgrades, EIP-equivalent proposals, and version transitions.
- **The attestation framework (§8.5).** Maintaining the federation of attestation issuers, audit standards, and revocation governance for hardware-attested mode.
- **The Liquid Media Chain (§6).** Operating the chain sequencer during bootstrap; coordinating the migration to decentralized stake-based sequencer consensus by a defined network-maturity milestone.
- **The LMA token.** Holding the initial token treasury, operating the periodic burn auctions (§11.8), maintaining the emission schedule for storage providers (§7) and threshold-network operators (§8.2), administering staking and slashing parameters.
- **Treasury.** Custody of the foundation’s operating allocation of LMA and the USDC operating carve-out from protocol-fee revenue; application to shared infrastructure costs and grants to ecosystem development.
- **Token-holder governance.** Per-protocol parameter changes (fee rates, emission curve adjustments, attestation federation milestones, chain parameter changes) by vote of staked LMA holders. During the bootstrap phase (Years 1–3), the foundation board retains veto power; sunset of board veto in favor of full token-holder governance is committed to a defined maturity threshold.

16.3 What the Foundation Does Not Do

The foundation does not operate any commercial storefront, does not manufacture or sell any reference Seed hardware, does not take fees on bilateral commercial relationships between ecosystem participants, does not hold any extractive position in LMA appreciation, and does not gatekeep participation in the standard-tier protocol operations. The hardware-attested mode’s federated framework (§8.5) is the only context in which the foundation acts as a gating party, and even there it acts as one of multiple federated issuers rather than as a sole authority.

16.4 The LMA Token

LMA is the native token of the Liquid Media Chain and the coordination instrument for storage-provider and threshold-network participation. Its three functions compose into a single internally consistent economic system: gas on the chain (Section 6); stake collateral for storage providers and threshold-network operators (§§7, 8); and value-capture instrument via burn from protocol fees (§11.8) and burn from chain base-fee mechanics.

The token's maximum supply is fixed at one billion LMA at launch, with allocation across the network-operator emission reserve, foundation operations, ecosystem development grants, early contributor pool, strategic partner reserve, foundation endowment, and public bootstrap reserve. The network-operator emission reserve releases via a decaying schedule modeled on Filecoin's storage reward emission¹⁷, with high early emission to attract bootstrap participation and asymptotic decay thereafter. Detailed parameters are specified in a companion token-economics document published alongside the foundation's incorporation.

A core structural property of the allocation: the commercial entity Wylloh, Inc. — the first commercial implementation of the protocol (§17) — receives zero LMA allocation. Individuals who contribute to the foundation's bootstrap may receive LMA grants as individuals against their foundation contributions, separate from any equity they hold in commercial implementations. This separation is the load-bearing structural commitment for the protocol's "non-extractive" framing: the foundation, the token, and the commercial implementations are kept in three distinct entity categories, and no commercial implementation depends on token economics for its revenue.

The token is designed and structured to function as a network utility instrument rather than as an investment instrument. Specifically: there is no initial coin offering or public token sale; distribution is via emission (for work performed), grants (for verified ecosystem contribution), and small structured strategic allocations; functional utility (gas, staking, governance) is operational at chain launch; no entity claims returns from LMA appreciation; the Cayman Foundation Company structure enables securities-compliant issuance under a regulatory framework with established precedent for protocol-utility tokens. These structural choices are intended to satisfy applicable securities analysis by ensuring that any reasonable acquirer of LMA acquires it for protocol utility rather than for profit from another entity's efforts.

Implementation status: the foundation is in formation; incorporation as a Cayman Foundation Company is a funded milestone of the upcoming raise. The Liquid Media Chain is in design; launch under foundation governance is a Year-1 milestone. The LMA token is specified; launch under foundation operation is concurrent with the chain launch. V5's permissionless minting and verified-publisher registry are specified and implemented in the reference contract scaffold; deployment under foundation governance is a 2026 milestone.

16.5 Curation: Gateless, Not Absent

The protocol layer is permissionless — any wallet can mint a title (V5 onward). Curation happens at the application and registry layers, not at the protocol. Three signals work together:

1. **Cryptographic identity at mint.** The wallet that mints a title is recorded as the publisher, on-chain, immutably. Brand identity is wallet-bound and cannot be impersonated.
2. **Foundation-maintained verified publishers registry.** The Liquid Media Foundation publishes a curated list of verified publishers at liquidmediafoundation.org/publishers.json. Verification signals quality and accountability without restricting the protocol; unverified publishers can still mint, but will not appear in verified-only application views.
3. **Application-layer editorial choice.** Each platform built on LMAP applies its own filter: the Wylloh storefront shows titles from Wylloh and other respected publishers; an indie aggregator might show all unverified content; a music-focused platform filters by `mediaType: music` and music-publisher wallets. The protocol exposes the truth (who minted what); applications shape the experience.

This model — *permissionless protocol, curated applications* — mirrors how DNS works: anyone can register a domain, but resolvers, browsers, and search engines each apply their own filters. Quality emerges economically rather than by gatekeeping. A publisher with poor taste mints work nobody buys; a publisher with good taste accumulates verified status and earns the publisher fee from real demand.

The legacy V4.1 registry retains its role-gated mechanism as the editorial mechanism for the legacy-construction deployment serving already-minted titles. New tokenizations under V5 land in the permissionless model.

17. Comparison with Prior Work

LMAP inherits ideas from several adjacent ecosystems and addresses limitations specific to each.

Centralized streaming services (Netflix, Disney+, and others) own the relationship with audiences and bear infrastructure costs proportional to viewership. They cannot offer durable digital ownership because durable ownership conflicts with subscription economics. LMAP inverts this by routing ownership through a public ledger and distribution through a peer-to-peer network.

Kaleidescape ships hardware-backed ownership for theatrical-quality content but operates within a closed ecosystem at luxury price points and serves a narrow demographic. LMAP shares the philosophical commitment to genuine ownership but extends the addressable audience by separating the protocol from any particular hardware vendor and accepting standard 4K HDR streaming quality rather than requiring lossless masters.

Plex demonstrates the “headless server at home plus client app on the television” architecture at scale (over 25 million users) with primarily user-ripped content of unclear provenance. LMAP-compatible Seed devices inherit this architecture and address Plex’s structural limitations: no provenance, no royalty flow, no curatorial focus, closed-source business model. The reference Seed devices extend Plex’s pattern with on-chain provenance and an open protocol.

Filecoin¹⁸ and **Helium** demonstrate decentralized-network plays with token-incentive mechanisms. Both attracted speculative participants more than utility-driven users, primarily because the hardware in each case had no standalone consumer value. The purchase rationale collapsed if the token economics did not deliver expected returns. LMAP’s reference Seed devices have primary consumer utility: the operator’s film library, accessible to LMAP-compatible playback clients on their local network. This utility does not depend on any token-incentive mechanism. Network participation is a beneficial side effect of a device that operators would buy regardless. This sidesteps the chicken-and-egg adoption problem that has limited prior decentralized-CDN plays.

Lit Protocol¹⁹ demonstrates threshold-mediated key release as a production-grade primitive for on-chain access control. LMAP incorporates Lit Protocol as the Year-0 bootstrap substrate for threshold release (§8.3); the Year-1 LMAP-native threshold network implements the same threshold-cryptography primitive operated by LMAP-staked nodes on the Liquid Media Chain. The relationship is explicit and time-bounded: bootstrap dependency, not permanent reliance.

Polygon and the Polygon CDK²⁰ provide the chain settlement substrate and the zero-knowledge rollup framework on which the Liquid Media Chain is constructed. LMAP is to Polygon what application-specific Layer-2 chains generally are to their Layer-1 anchors: a domain-specific execution environment that benefits from the security and liquidity of the underlying chain while tuning block economics to the application’s requirements.

Bitcoin²¹ and **Ethereum**²² establish the cryptographic and economic primitives this protocol relies upon. The protocol’s contributions specific to film distribution are the dual-contract architecture (§3), the modular rights-stacking mechanism (§4), the threshold-mediated access control model (§8), the permanence guarantee (§9), the Seed-network distribution model (§10), the publisher-author distinction with royalty-shareholder distribution (§11.2–§11.3), and the LMA token’s burn-from-fee value-capture mechanism (§11.8).

18. Conclusion

We have proposed a protocol for film distribution addressing fundamental misalignments in centralized streaming. By tokenizing films into standardized units with embedded licensing rights, we create genuine digital ownership. By distributing content through a peer-to-peer network growing stronger with demand, we invert economics that punish platforms for success. By automating royalty distribution including on secondary sales, we ensure transparent creator compensation through markets that never existed for physical media.

Threshold-mediated key release replaces the legacy derivable-key construction as the production cryptographic model. Access to a film’s decryption material is gated by current on-chain ownership at decryption time, evaluated by a threshold network rather than by any single party. The protocol claims, and intends to demonstrate, security equivalent to or stronger than legacy hardware-attested DRM. Forward-compatible support for hardware-attested key wrapping exists as a capability for licensing relationships predicated on legacy industry frameworks, governed by a federated issuance framework structurally independent from any commercial operator; we expect those frameworks to evolve toward recognizing threshold cryptography directly, and the protocol’s center of gravity is positioned accordingly.

The separation of distribution tokens from copyright ownership enables fluid rights markets while preserving creator control over intellectual property. Commercial exhibitors can acquire rights directly from collectors. Studios can acquire copyright for derivative works without disrupting existing token markets. Rights flow to natural owners through market mechanisms rather than negotiated deals.

A dedicated Layer-2 chain, a single coordination token, an independent foundation, and a federated attestation framework together produce a protocol whose external dependencies sunset over a defined three-year horizon. The protocol’s intellectual integrity does not require any commercial entity’s continued existence; the chain remembers, the network distributes, the contracts execute.

The protocol requires no permission from existing gatekeepers. Public-domain content bootstraps the network, demonstrating capabilities before commercial titles join. As the network grows, established platforms face a choice: implement the protocol and benefit from its efficiencies, or cede ground to peer marketplaces and competitors that do.

LMAP represents not disruption but recoherence — using peer-to-peer technology once deployed to undermine the industry to instead restore fair compensation and sustainable economics. The same distributed networks that enabled unauthorized copying can now enable legitimate, liquid, permanent ownership. *In the venom, the antidote.*

19. Bootstrap Roadmap

The protocol’s external dependencies are explicitly bootstrap-only. Each is replaced by an LMAP-native substrate on a defined timeline.

Year 0 — Bootstrap. The protocol operates on Polygon mainnet with the V4.1 registry contract. Threshold-mediated key release runs against Lit Protocol’s Naga mainnet. Content storage uses IPFS with a centralized pinning service operated by the first commercial implementation as a high-availability substrate. The foundation is in formation; the Liquid Media Chain is in design; LMA does not yet exist on-chain.

Year 1 — Native Substrate Launch. The Liquid Media Foundation incorporates as a Cayman Foundation Company. The Liquid Media Chain launches as a Polygon CDK zero-knowledge rollup with LMA as native gas. LMAP V5 contracts deploy natively on the chain. The LMA token launches under foundation operation. The LMAP-native threshold network deploys with foundation-operated bootstrap nodes plus permissionless stake-based participation. The storage-provider registry and proof-of-retrievability mechanism activate. New content defaults to the LMAP-native threshold network increasingly through the year. Periodic LMA burn auctions begin operation.

Year 2 — Dependency Sunset. All production content has migrated to the LMAP-native threshold network. The Lit Protocol dependency is retired. The centralized pinning fallback is decommissioned. The attestation framework’s first federation milestone is achieved (at least one independent issuer beyond the foundation co-signs attestation credentials). The Liquid Media Chain sequencer begins decentralization via stake-based consensus.

Year 3 — Mature Operation. Token-holder governance replaces foundation-board veto for protocol parameter changes. The attestation framework operates with multiple federated issuers. The chain operates with decentralized sequencer consensus. The protocol’s external dependencies are fully resolved; the protocol operates self-sufficiently

on its own substrate.

This roadmap is a public commitment, not a roadmap of intent. Each milestone is a defined deliverable with a published acceptance test; the foundation's annual report enumerates progress against each.

Implementation Status Summary

A precise summary of the protocol's components by their status as of this paper's publication:

- **Shipped (legacy V4.1 deployment on Polygon mainnet):** ERC-1155 distribution registry at `0x8e834c6031A2d59e1cADd24Ab24F668301ed7c1D`; role-gated minting for the bootstrap deployment; modular rights stacking; chunked AES-256-GCM encryption; legacy deterministic key derivation (now scoped to public-domain demonstration content, not for new production); self-contained manifest reading from on-chain metadata + IPFS, with storage-service fallback; open-source reference Seed daemon (verify ownership, fetch + decrypt, auto-pin, LAN-stream over HTTP/mDNS); web client at `wylloh.com`; first tokenized film (*The Cocoanuts*, 1929) sold and downloaded end-to-end across two independent architectures (macOS arm64, Raspberry Pi aarch64) with byte-identical output. The V4.1 contract continues to serve existing tokens.
- **In active migration (Year-0 bootstrap):** migration of the Wylloh reference web client and Seed daemon from the legacy derivable-key construction to threshold-mediated key release via Lit Protocol's Naga mainnet. Completion of this migration is a blocking prerequisite for tokenizing any commercial content; the V4.1 deployment continues to serve already-tokenized public-domain content under the legacy construction during and after the migration.
- **Specified, in active implementation (V5 on Polygon mainnet during Year 0; on Liquid Media Chain from Year 1):** the LMAPRegistryV5 reference contract under foundation governance — adds the three-way revenue split (§11.1), publisher/author distinction (§11.2), royalty shareholder distribution (§11.3), atomic batch acquisition for rights stacking (§11.5), `mediaType` field for media-agnostic operation, updateable platform treasury via multi-sig timelock, ERC-4337 paymaster hooks, and permissionless minting (§16.5). Scaffold with passing tests; full marketplace implementation, comprehensive test coverage, and deployment under foundation governance are 2026 milestones.
- **Specified, foundation-formation work (Year 0):** Cayman Foundation Company incorporation; securities-counsel review of the LMA token structure; constitution of the foundation board; publication of the verified-publishers registry; constitution of the first attestation issuer.
- **Specified, Year-1 deployment:** the Liquid Media Chain as a Polygon CDK zero-knowledge rollup; LMA token launch; LMAP-native threshold network with stake-based participation; the storage-provider registry and proof-of-retrievability mechanism; periodic LMA burn auctions; LMAP V5 contracts on the chain natively; bridge contracts to Polygon mainnet for USDC and legacy V4.1 holdings.
- **Specified, Year-2+ deployment:** Lit Protocol dependency sunset; centralized pinning sunset; first attestation-issuer federation milestone; sequencer decentralization to stake-based consensus.
- **Specified, future protocol versions (V6+):** ERC-721 copyright registry; on-chain staking for commercial-exhibition windows; presale-funding milestone-escrow contracts; threshold dispersal of ciphertext shards; the hardware-attested mode with **normatively-required hardware-attested per-device key wrapping** (§8.4). The wire-level attestation specification (report format, credential schema, issuer federation protocol, revocation flow) is committed to a separate LMAP V6 technical specification, forthcoming.
- **Long-horizon (operational maturity required):** studio-grade licensing through hardware-attested mode with major rights holders; full token-holder governance of protocol parameters with foundation-board veto retired; decentralized chain-sequencer consensus operating at production load.

This paper is a working specification. Components shipped today behave as described; components specified are commitments the protocol is engineered toward. The Liquid Media Foundation, once formally established, will maintain this status summary as a living public artifact.

Changelog

Version 3.0 (June 2026). Substantial revision following the V5 contract-specification work and the architectural decisions of the seed-round preparation. Four structural changes: (a) the legacy *open tier* of the §7 cryptography framing is retired as a production model — the deterministic key-derivation construction is reclassified as a public-domain demonstration approach, valid for the V4.1 deployment of *The Cocoanuts* but not appropriate for new commercial content; threshold-mediated key release becomes the production security model. (b) The protocol claims and intends to demonstrate security equivalent to or stronger than legacy hardware-attested DRM through threshold cryptography with stake-based sybil resistance; hardware attestation is reframed as forward-compatibility for legacy industry licensing frameworks (§8.4) rather than as a co-equal “mode” of the security architecture. The expectation is that industry compliance frameworks evolve to recognize threshold cryptography directly; the hardware-attested capability exists at the periphery of the protocol’s design, decreasing in significance over time. (c) A dedicated Layer-2 chain — the *Liquid Media Chain*, §6 — is introduced as the protocol’s operational substrate, hosting the V5 contracts and using LMA as native gas. (d) A single coordination token — *LMA* — is introduced (§16.4) for storage incentives (§7.1), threshold-network staking (§8.2), gas, and value-capture via burn from protocol fees (§11.8). The foundation’s legal structure is specified as a Cayman Foundation Company (§16.1), with the explicit caveat that this commitment precedes formal securities-counsel review attendant to LMA token issuance and may be revised in v3.1 should counsel recommend a structurally superior alternative. The implementation-status summary is expanded to reflect bootstrap, migration, and native-substrate-launch phases. The closing line of v2.3 — *In the venom, the antidote* — is preserved.

Version 2.3 (May 2026). Reference Seed daemon promoted from “specified” to “shipped” in the Implementation Status. §9 (Seed Network) updated to reflect end-to-end operation of the open-source reference daemon on independent ARM64 hardware (Raspberry Pi 4) with byte-identical decrypted output across architectures and no dependency on any centralized service. §7.2 (Certified Tier) gained a normative statement on hardware binding. A clarifying paragraph distinguished *playback* (which requires certified hardware) from *carriage* (which does not — tokens, encrypted content, and pinning remain freely accessible across all hardware).

Version 2.2 (May 2026). V5 reference contract specification additions. §10 (Marketplace Mechanics) substantially expanded to specify the V5 economic model: three-way revenue split, distinct publisher/author identities, royalty shareholder distribution supporting up to 50 collaborators per title, atomic batch acquisition for rights stacking, and an honest articulation of the royalty enforcement boundary. §15 (Governance) updated to reflect V5’s permissionless minting model.

Version 2.1 (May 2026). Rebrand pass under Liquid Media Foundation stewardship. The protocol — previously referred to internally as “Wylloh” — is renamed *LMA* (Liquid Media Access Protocol) to formalize the separation between the open standard (LMA, stewarded by the foundation) and the first commercial implementation (Wylloh, the company building the reference hardware and storefront).

Version 2.0 (April 2026). Comprehensive revision following the April 2026 mainnet launch and the architectural pivot to the headless-Seed network. Key additions: layered architecture, two-tier attestation model (now superseded by §8’s threshold-mediated release with optional hardware-attested mode), permanence guarantees, governance and the foundation, comparison with prior work.

Version 1.0 (April 2026). Initial publication.

References

1. W. Radoski et al., “EIP-1155: Multi Token Standard,” Ethereum Improvement Proposals, 2018.
2. W. Entriiken et al., “EIP-721: Non-Fungible Token Standard,” Ethereum Improvement Proposals, 2018.
3. Polygon Labs, “Polygon CDK: A Framework for Sovereign ZK-Rollups,” docs.polygon.technology, current edition.
4. V. Buterin et al., “ERC-4337: Account Abstraction Using Alt Mempool,” Ethereum Improvement Proposals, 2021.
5. T. Beiko, A. Davidson, V. Buterin et al., “EIP-1559: Fee Market Change for ETH 1.0 Chain,” Ethereum Improvement Proposals, 2019.
6. J. Benet, “IPFS — Content Addressed, Versioned, P2P File System,” arXiv:1407.3561, 2014.
7. Protocol Labs, “Filecoin: A Decentralized Storage Network,” filecoin.io technical specification.
8. A. Shamir, “How to Share a Secret,” *Communications of the ACM*, vol. 22, no. 11, pp. 612–613, 1979.
9. W. Stewart, J. Galindo, “EIP-4361: Sign-In with Ethereum,” Ethereum Improvement Proposals, 2021.
10. D. Hyun et al., “Lit Protocol: Decentralized Access Control with Threshold Cryptography,” developer.litprotocol.com, current edition.
11. H. Krawczyk and P. Eronen, “HKDF: HMAC-based Extract-and-Expand Key Derivation Function,” IETF RFC 5869, May 2010.
12. ARM Limited, “ARM TrustZone Technology,” developer.arm.com, current edition.
13. Trusted Computing Group, “TPM 2.0 Library Specification,” trustedcomputinggroup.org, current edition.
14. H. Birkholz et al., “Remote Attestation procedureS (RATS) Architecture,” IETF RFC 9334, January 2023.
15. V. Buterin et al., “ERC-4337: Account Abstraction Using Alt Mempool,” Ethereum Improvement Proposals, 2021.
16. T. Beiko, A. Davidson, V. Buterin et al., “EIP-1559: Fee Market Change for ETH 1.0 Chain,” Ethereum Improvement Proposals, 2019.
17. Protocol Labs, “Filecoin: A Decentralized Storage Network,” filecoin.io technical specification.
18. Protocol Labs, “Filecoin: A Decentralized Storage Network,” filecoin.io technical specification.
19. D. Hyun et al., “Lit Protocol: Decentralized Access Control with Threshold Cryptography,” developer.litprotocol.com, current edition.
20. Polygon Labs, “Polygon CDK: A Framework for Sovereign ZK-Rollups,” docs.polygon.technology, current edition.
21. S. Nakamoto, “Bitcoin: A Peer-to-Peer Electronic Cash System,” 2008.
22. V. Buterin, “Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform,” 2014.